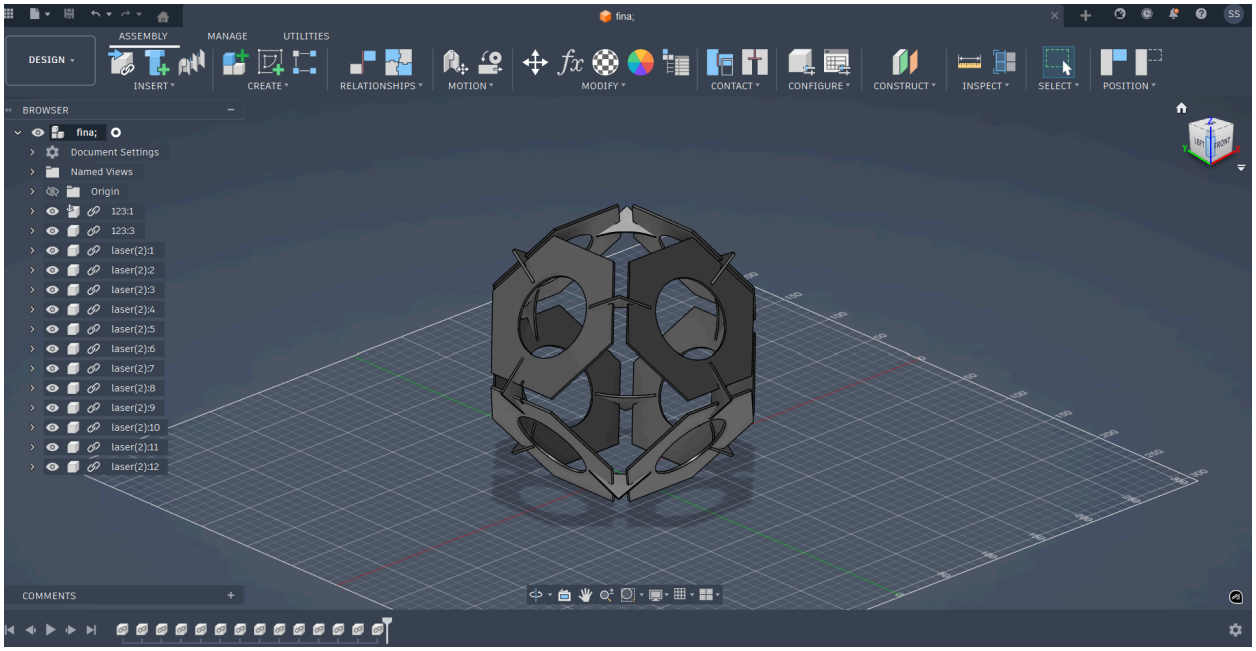
Laser Cut Geometric Assembly – Fusion 360 Design Explanation



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1. Introduction

This project is a **laser-cut interlocking geometric structure** designed in Fusion 360.
It is a **modular slot-fit assembly** made from multiple identical panels arranged to form a spherical / polyhedral structure.

This model demonstrates:

- Parametric design
- Assembly planning
- Slot fitting mechanism
- Manufacturing-ready laser cutting concept

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2. Design Overview

From the image, the structure consists of:

- Multiple identical polygonal panels
- Circular cut-outs in each face
- Interlocking slots on edges
- Symmetrical 3D assembly
- No screws or adhesives (self-locking design)

Each panel is designed to slide into another using **precisely measured slots**, forming a stable 3D object.

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3. Unique Features of My Design

🔷 1. Slot-Fit Mechanism

- Each panel has rectangular slots.
- The slot width equals material thickness.
- Ensures tight friction fit after laser cutting.

🔷 2. Circular Cut-outs

- Large circular openings reduce material weight.
- Improves aesthetics.
- Maintains structural balance.

🔷 3. Symmetrical Assembly

- All faces are evenly distributed.
- The structure maintains geometric balance.
- Good example of spatial design.

🔷 4. Manufacturing-Oriented Design

- Designed specifically for:
    - MDF
    - Acrylic
    - Plywood
- Easy to fabricate using laser cutting machine.

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4. Tools Used in Fusion 360

- Sketch tool – To create polygon and circle profiles
- Extrude – To give thickness
- Pattern tool – To duplicate panels
- Slot design – For interlocking joints
- Assembly (Joint) – To position panels
- Parametric constraints – For dimensional accuracy

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5. Design Process (Step-by-Step)

1. Created base polygon sketch.
2. Added circular cut-out inside panel.
3. Designed rectangular slots on edges.
4. Extruded the panel to required thickness.
5. Converted into component.
6. Used circular pattern / copy to create multiple panels.
7. Assembled panels using joint alignment.
8. Checked clearances and fit.

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6. Engineering Concepts Demonstrated

✔ Parametric modeling
✔ Tolerance planning
✔ Press-fit design
✔ Laser manufacturing understanding
✔ 3D spatial reasoning

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7. Applications

This type of structure can be used for:

- Decorative lamp covers
- Exhibition models
- Structural prototypes
- Architectural miniatures
- STEM educational models

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8. Why This Design is Advanced

This is not a simple 3D model. It requires:

- Accurate slot dimension calculation
- Assembly visualization before manufacturing
- Understanding of material thickness
- Geometric symmetry planning

It shows strong CAD and product design skills.